ANTALYA BİLİM UNIVERSITY
Computer Engineering Department

Cybershield

“ Multi-Threat AI-Enhanced Firewall with Explainable Blocking and GeoIP Intelligence ”

This project is a custom-built firewall that uses AI and behavioral techniques to detect and block phishing, ransomware, malware, and DDoS attacks.

We used:

Python

Kali Linux

nftables

Scapy

Flask

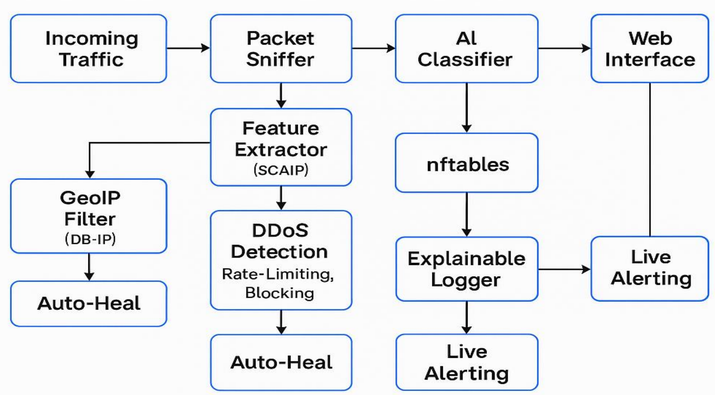
SQLite

DB-IP GeoIP Database

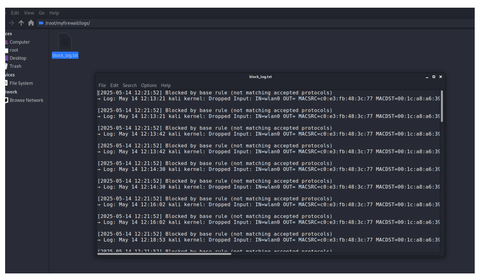
Our Goals&Features:

➡Explainable Blocking (log reasons)
➡Auto-Heal for broken rules
➡GeoIP Watch + Auto-Ban
➡DDoS detection (log-based)
➡Scapy-based packet inspection

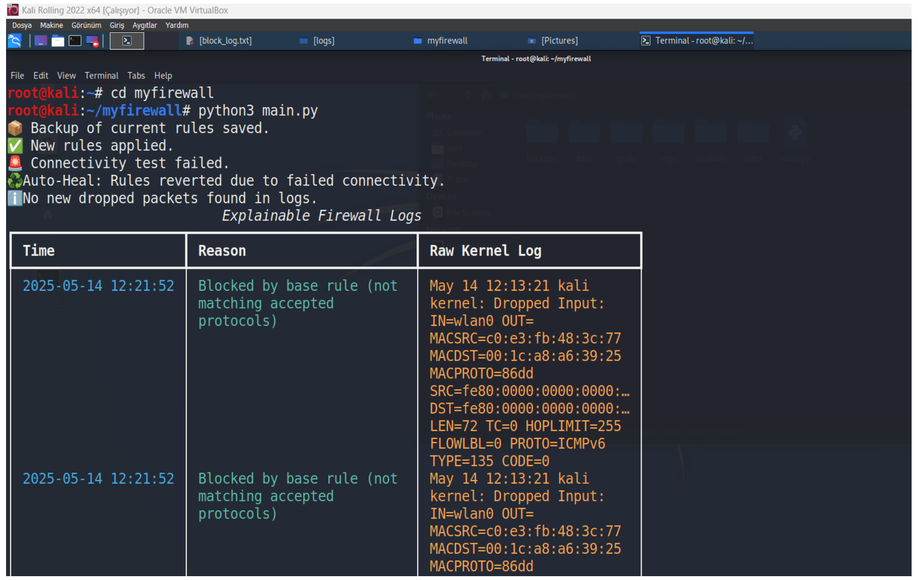
System Architecture Diagram



Log file screenshot



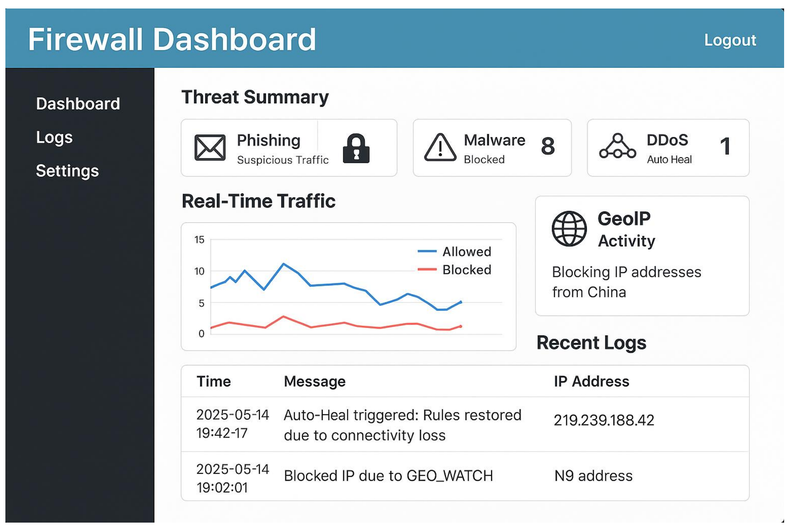
This screenshot shows how the firewall system applies a rule, checks connectivity, and reverts automatically if a misconfiguration is detected.



Real-Time Logs and Auto-Heal Feedback

Our unique features:
-Auto-Heal Mechanism for Misconfigured Rules
-Explainable Blocking

➡ This firewall can be deployed on Raspberry Pi, home routers, or Linux gateways to provide smart, adaptive security for small networks or IoT.



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